



HydroGraphRAG: Intelligent GIS Assistant

Geospatial code generation and visualization based on knowledge graphs.

Enter hydrology query:

Уровень воды в реке Или

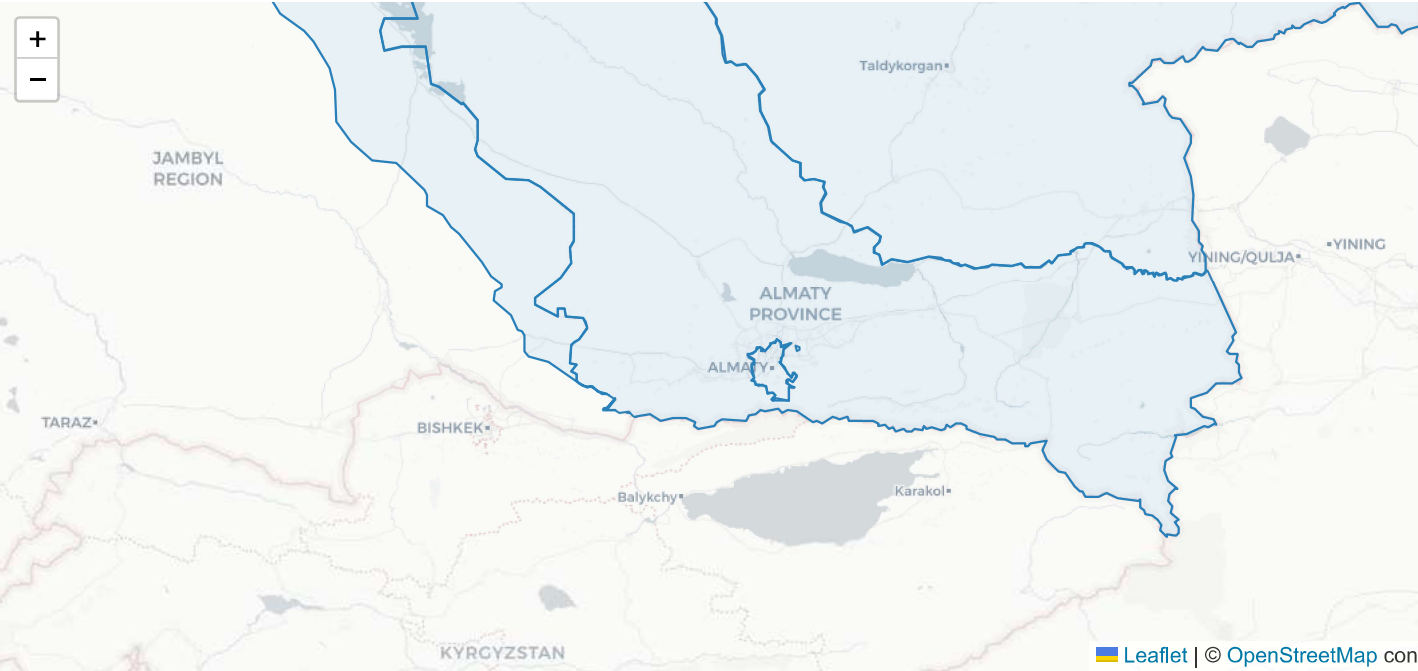
Generate solution

 Solution generated!

Category	Triples extracted	Points on map
EXPLICIT	2	0



Geospatial Visualization





Analytical Report and Code

Modeling Solution:

Unfortunately, the graph database is missing data.

Implementation Code:

```
import geopandas as gpd
import folium
from shapely import wkt
```

```
# 1. Load basin
basin_data = gpd.read_file(r"data/basin_data.shp").to_crs('EPSG:4326')
centroid = basin_data.geometry.centroid.iloc[0]

# 2. Init map
m = folium.Map(location=[centroid.y, centroid.x], tiles='CartoDB positron', zoom_start=8)
folium.GeoJson(basin_data.to_json(), style_function=lambda x: {'fillColor': 'green', 'color': 'darkgreen', 'fillOpacity': 0.5})

# 3. Add YOUR points here based on Graph Knowledge
points = [
    # Example: {"name": "River Name", "wkt": "POINT(76.0 43.0)"}
]

# 4. Draw markers
for p in points:
    geom = wkt.loads(p["wkt"])
    folium.Marker(location=[geom.y, geom.x], popup=p["name"]).add_to(m)

m.save("api_1780739271.html")
```